Joseph Mills

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SKILLS

Languages: Python, Java, Go, TypeScript, JavaScript, SQL, C++, Rust, C, Haskell, R, C#
Machine Learning, AI & Data: PyTorch, ONNX, TensorFlow Lite, LangGraph, Gemini, RAG, Qdrant, pandas, GeoPandas, SciPy, Power BI, Databricks
Web & Backend: React, Next.js, Node.js, Express, FastAPI, .NET, ASP.NET Core, REST APIs, Postgres, SQLite, Redis
DevOps & Tools: Docker, CI/CD, GitHub Actions, Linux, Git, PowerShell, Playwright, AWS, CDK, DynamoDB, Azure, Vercel, Supabase, Kubernetes, Agile/Scrum

EDUCATION

Mount Royal University Sep. 2023 – Apr. 2027
Bachelor of Science in Computer Science | CGPA 3.59 *Calgary, AB*

- Awards: NSERC Undergraduate Student Research Award (USRA), Dean's Honour Roll
- Relevant coursework: Artificial Intelligence, Advanced Software Engineering, Algorithms and Complexity, Data Structures (**Java**), Operating Systems, Databases, Network Infrastructure, Advanced Web Development, Programming Paradigms, Statistics, Linear Algebra for Data Science

EXPERIENCE

Software Engineer and Machine Learning Research Assistant Jan. 2026 – Present
Mount Royal University *Calgary, AB*

- Built SplitBench, an open-source **Python** split federated learning benchmark comparing 7 methods on 7 datasets
- Co-authored a paper accepted at IEEE IRI 2026 and took part in code reviews with the research supervisor
- Predicted training time within 2% with a **Python** scheduler that splits work by device and network speed
- Automated 80+ unattended **PowerShell** runs across 27 **Docker** setups simulating slow devices and networks
- Nearly doubled rare-class F1 on medical images (28% → 52%) using **PyTorch**
- Reached 87.6% accuracy on a 7-class medical image dataset by testing 20 **PyTorch** model setups

Data Science Intern Jan. 2026 – Mar. 2026
Miistakis Institute *Calgary, AB*

- Fixed mapping errors across a 260,000-polygon geodatabase with **GeoPandas**, aligning multi-source data
- Cut a multi-day Excel process to minutes with a **Python** pipeline, after interviewing staff to find it
- Briefed external stakeholders with a report turning **SciPy** biodiversity statistics into plain-language findings

Jump Crew and Signage Team May 2023 – Sep. 2025
Spruce Meadows *Calgary, AB*

- Completed full course changes within 5-minute commercial breaks at live televised events by coordinating the crew
- Planned weekly signage work with upper management and trained the next season's summer manager

PROJECTS

Boring Browser | *C++, Rust, Python, Chromium, GitHub Actions* jd-d3v.github.io/boring-web-browser | 2026

- Shipped 2 public releases of a privacy-focused **Chromium** browser built on 37 **C++** source changes
- Chose safety features by interviewing scam victims and surveying Reddit users
- Integrated a **Rust** ad-blocking engine that blocks trackers disguised as first-party content
- Automated signed releases and auto-updates with a **GitHub Actions** CI/CD pipeline
- Tested every build with 30+ **Python** end-to-end suites covering every major feature

Frostline | *Go, AWS, C++, C#/.NET, Next.js* 2026

- Built a pump-house freeze alarm that texts, then phones family in order until someone presses 1
- Designed an event-driven **Go** backend on **AWS** Lambda, Step Functions and IoT Core, deployed with CDK
- Wrote **C++** firmware for an ESP32 LTE-M sensor box that buffers 50 hours of readings through signal loss
- Built a **C#** API and **Next.js** dashboard with passwordless sign-in and per-farm access control
- Tested with 1,300+ automated tests and 6 CI jobs including **Playwright** and API contract checks

Pipe Dreams | *Python, React, TypeScript, FastAPI, scikit-learn, GeoPandas, ElevenLabs* 2026

- Won Best Project Built with **ElevenLabs** at IEEE Southern Alberta Hackathon 2026 (26 teams), team of 2

- Built the **React** dashboard and **FastAPI** backend for an AI agent ranking water mains from 37,538 breaks
- Added an **ElevenLabs** voice agent that explains each pipe's ranking to city staff

Beyonder | *Python, LangGraph, Gemini, FastAPI, Next.js, Postgres, Qdrant, Docker* jd-d3v.github.io/beyonder | 2026

- Built a web-novel translation app with a self-correcting **LangGraph** agent on **Gemini**
- Answered reader questions spoiler-free with RAG over only finished chapters (**Qdrant** semantic search)
- Fixed long-chapter timeouts with resumable chunked translation, fitting the app on a 512 MB server

Woofer | *JavaScript, TypeScript, Go, Next.js, Postgres* woofer-web-ashen.vercel.app | 2026

- Deployed a dog adoption app in **Next.js** on **Vercel** where adopters browse dogs and shelters review applicants
- Designed an authenticated **Go** REST API on **Postgres** for listings, matches, and messaging

Classroom Finder | *React, TypeScript, Python, Playwright* jd-d3v.github.io/Class-Finder | 2025

- Built a **React**/**TypeScript** web app and Telegram bot showing free classrooms from the university timetable
- Wrote the **Python** REST backend serving both apps, matching room requests against class times

Music Browser | *JavaScript, React, Node.js, Express, SQLite* jd-d3v.github.io/COMP4513Assign2 | 2026

- Deployed a music catalogue app in **React** with a **Node.js**/**Express** REST API on **SQLite**
- Recommended the 3 most similar tracks per song and charted 6 audio measurements with filtering and sorting

Multi-Threaded Web Server | *C, POSIX Threads, Sockets, Linux* github.com/JSenk347/web-server | 2025

- Served connections in parallel across 4 workers with a **C** thread pool and **pthreads** producer-consumer queue
- Added load shedding for full queues and a live dashboard of queue size and total requests (team of 3)

ATHLETICS AND ACTIVITIES

Biathlon, Team Alberta 2013 – 2020
Foothills Nordic *Calgary, AB*

- Won gold at the North American Championships and bronze at the Canadian National Championships
- Selected to Team Alberta; attended national team camps and performance testing with national team athletes
- Trained 11 hours per week for 5 years while completing high school full-time

Biathlon Coach 2019 – 2023
Bragg Creek Nordic *Bragg Creek, AB*

- Coached junior athletes for 4 seasons, running training, range safety, and race-day support
- Prepared new athletes for their first provincial races

MRUHacks Hackathon Oct. 2025
Mount Royal University *Calgary, AB*

- Built MoJo, a peer-to-peer payment app, in 24 hours as a team of 5 (React Native, Flask, MongoDB, AWS)